

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Medium

Question

North American Thrasher Mean Bill Size and Habitat Temperature Range

Species	Mean bill surface area (cm ²)	Mean maximum temperature of warmest month (°C)	Mean minimum temperature of coldest month (°C)
Brown thrasher	1.86	30.40	-4.29
Bendire's thrasher	1.98	36.57	0.24
Long-billed thrasher	2.24	35.27	8.82
Cozumel thrasher	2.28	33.27	18.21
Ocellated thrasher	3.26	27.56	5.45

It has been hypothesized that since birds can dissipate excess heat through their bills, bill size should increase with habitat temperature. To evaluate this hypothesis for a 2021 study, Charlotte Probst and colleagues gathered data on mean bill surface area of species of North American thrashers (genus: *Toxostoma*) as well as on climate conditions of the birds’ native habitats. Based on their data, Probst and colleagues concluded that the hypothesis was not fully supported.

Which choice most effectively uses data from the table to support Probst and colleagues’ conclusion?

Answer

- A. Although the Bendire’s thrasher has one of the smallest mean bill surface areas of the birds included in the table, its habitat has one of the lowest mean maximum temperatures in the warmest month and one of the lowest mean minimum temperatures in the coldest month.
- B. Although the Cozumel thrasher has the second greatest mean bill surface area of the birds included in the table, its habitat’s mean temperature in the warmest month is significantly higher than that of the other birds’ habitats.
- C. Of the birds included in the table, the brown thrasher has the smallest mean bill surface area and the habitat with the lowest mean minimum temperature in the coldest month, while the long-billed thrasher has the second-largest mean bill surface area and the habitat with the second-highest mean minimum temperature in the coldest month.
- D. Of the birds included in the table, the ocellated thrasher has the largest mean bill surface area and the habitat with the lowest mean maximum temperature in the warmest month, while the Bendire’s thrasher has the second smallest mean bill surface area and the habitat with the highest mean maximum temperature in the warmest month.

Correct Answer: D

Rationale

Choice D is the best answer because it most effectively uses data from the table to support Probst and colleagues’ conclusion about the hypothesis regarding the relationship between birds’ bill size and habitat temperature. The text states that although it has been hypothesized that birds’ bill size should increase with habitat temperature (an adaptation that reflects the fact that birds can release excess heat through their bills), Probst and colleagues concluded after studying mean bill surface area in North American thrashers that this hypothesis was not fully supported. In other words, Probst and colleagues did not observe the predicted positive relationship between mean bill surface area and habitat temperature in the thrashers they studied. The data in the table about the ocellated thrasher support Probst and colleagues’ conclusion because

this species has the greatest mean bill surface area (3.26 cm^2), but its habitat has the lowest mean maximum temperature in the warmest month (27.56°C). If the hypothesis were fully supported, species such as the ocellated thrasher that have colder habitats would likely also have smaller bills. Similarly, the data about the Bendire's thrasher support Probst and colleagues' conclusion because this species has the second-smallest mean bill surface area (1.98 cm^2), yet its habitat has the highest mean maximum temperature in the warmest month (36.57°C). If the hypothesis were fully supported, the Bendire's thrasher and other species with warmer habitats would also have larger bills. Because data in the table show that the ocellated thrasher and the Bendire's thrasher have the opposite or near opposite of the hypothesized relationship between bill size and habitat temperature, these examples support Probst and colleagues' conclusion that this hypothesis isn't fully supported.

Choice A is incorrect because it misrepresents the data in the table. Although the Bendire's thrasher has the second-smallest mean bill surface area of the species that are included in the table and although its habitat has the second-lowest mean minimum temperature in the coldest month, this habitat has the highest (rather than the lowest) mean maximum temperature in the warmest month. Moreover, the data about this species' bill surface area and mean minimum temperature suggest that smaller bills are associated with colder habitats, supporting rather than challenging the hypothesis that bill size increases with habitat temperature. Choice B is incorrect because it misrepresents the data in the table. Although the Cozumel thrasher has the second-greatest mean bill surface area of the species that are included in the table, its habitat's mean maximum temperature in the warmest month is not "significantly higher" than the mean maximum temperatures of the other habitats. On the contrary, the mean maximum temperature of the Cozumel thrasher's habitat is lower than that observed for the habitats of both the Bendire's thrasher and the long-billed thrasher. Choice C is incorrect because it misrepresents the data in the table. Although this choice accurately describes the brown thrasher's mean bill surface area and habitat temperature, it incorrectly states that the long-billed thrasher has the second-largest (rather than the third-largest) mean bill surface area of the species included. In addition, instead of revealing a lack of support for the hypothesis that bill size increases with habitat temperature, the data in this choice broadly reflect the hypothesized relationship.